RAMCO INSTITUTE OF TECHNOLOGY
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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: VI semester B.E. ECE B
Course Code & Title: CS3491 Artificial Intelligence and Machine Learning
Name of the Faculty member (s): P. Venkatesh

Innovative Practice Description

- **Unit / Topic:** Unit IV / Expectation Maximization
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** 4c
- **Activity Chosen:** Flipped Classroom
- **Justification:**

It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning. Also the practice was chosen because the topic is simpler and interesting for the students to refer and learn by themselves

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

Specific topic was given to the students learn on their own. Resources like reference book, videos were given to the students. Apart from this they can refer any other relevant source for preparation. Students are asked to prepare more in depth than before. Students are separated into groups where students are given a specific topic to prepare and present. Get the class back together to share the individual group's work with everyone.

- **CO – PO / PSO mapping:**

CO	PO1	PO3	PO8	PO9	PO10
CO5	3	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO3	PO8	PO9	PO10
	3	2	1	1	1
Justification for	Solve real time	PO3 is moderately mapped with	Identify tenets of the	It helps the students to	The students will present the

correlation	problems in topics like Expectation Maximization and hence PO1 has been mapped strongly to level 3	level 2 because students will build new system using the Expectation Maximization algorithms	professional code of ethics while doing activity based learning. So, it is mapped at level 2	present the topic of discussion as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2	results and conclusions of the activity carried out in front of the class which require good communication and presentation skills and so PO 10 is mapped

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
    - ***Feedback of practice from students and other stakeholders:***
        - Students feel that they have improved self-learning.
        - They learn how to communicate with team members and work together.
        - They were completely involved
    - ***Benefit of the practice:***
        - Every student got equal opportunity to come forward to take part in this activity.
        - They felt easy in solving level 3 questions in Assignments and Internal Assessment test
        - The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct
    - ***Challenges faced in implementation:***

- Few students hesitated to come forward to present the results and conclusions of the disscussion
- The main challenge faced is that few students not exposed to flipped class room.

References:

- https://www.teachthought.com/learning/a-flipped-classroom/
- Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.

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INNOVATIVE TEACHING METHODS

Degree, Semester & Branch : VI Semester B.E. ECE B

Course Code & Title : CS3491 Artificial Intelligence and Machine Learning

Name of the Faculty member : P.Venkatesh

Topic: Adversarial Search